

Self-produced Pendulum: Numerical Data and Relationships

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Introduction:

Since the Newtonian era, pendulums are prime examples of systems that exhibit harmonic motion¹ (Britannica, 2008). The purpose of this lab is to determine the relationship between **Equation 1**, the equation of a damped harmonic oscillator² (Wilson, 2024) and my own homemade pendulum. It should be noted that uncertainties play a key role throughout the data collection, analysis and the conclusion drawn.

$$\theta(t) = \theta_0 e^{-t/\tau} \cos(2\pi \frac{t}{T} + \Phi_0) \quad (1)$$

The equation of the damped harmonic oscillator model, it takes into account the effects of “friction” on a pendulum and gradually “damps” the motion as time increases³ (MIT, 2017). Where θ_0 (rad) is the initial release angle; t (sec) is time elapsed; τ (sec) is a constant of the pendulum that measures frictional force; Φ_0 (rad) is the phase constant that accounts for time delay; T (seconds) is the period of the pendulum, i.e. how long does it take to undergo one cycle.

The lab setup of the pendulum includes a lock as a simple mass, a wooden block as the main body of the pendulum and fishing wire as the cable connecting the main body and the mass. For more details, refer to **Section A** of the report.

The purpose of **Part I.** is to determine a.) Whether or not the period of the pendulum depends on the initial release angle; and b.) The Q factor with two separate methods outlined and how do they compare.

Through experimental data, the following major conclusions were drawn:

1. The period of the pendulum depends on the initial release angle. This period is around constant only at angles $<|0.3|$ rad. This is determined by fitting the data into **Equation 2**.
2. The Q factor measured is different with “tau factor method” and “counting method”, those being 570 ± 30 and 620 ± 10 respectively, **Part II** of the lab will apply the “counting method”. To determine Q factor with the “tau factor method”, **Equation 3,4** is applied.

$$T = T_0(1 + B\theta_0 + C\theta_0^2) \quad (2)$$

Quadratic function that fits the initial angle to provide a quadratic relationship of the period². Where T (sec) is the period; T_0 (sec), B and C (unitless) are parameters for fitting; θ_0 (rad) is the initial release angle.

$$\theta = ae^{-p(t-d)} + c \quad (3)$$

*Exponential function that fits the Local Maximums (Amplitudes) from **Equation 1** to determine the constant tau. Refer to **Subsection aii** for more details. Where p (1/sec), d (sec), a and c (unitless) are parameters of best fit; e is Euler's constant ~ 2.718 ; θ is the amplitude of the pendulum, i.e. maximum angle it reaches in one cycle.*

$$Q = \pi \frac{\tau}{T} \quad (4)$$

*Equation that defines the Q factor being positively proportional to tau and inversely proportional to period². Where Q (unitless) is the “Quality factor” of the pendulum that measures frictional force; T (sec) is the period and τ (sec) is a constant (refer to **Equation 1**)*

Similarly, the purpose of **Part II.** is to determine a.) Whether or not a leading coefficient of 2 and a root function describe the relationship between the Period and the Length of the pendulum; and b.) The relationship (if any) between the Q Factor and the length of the pendulum.

Through experimental data, the following major conclusions were drawn:

1. When Period and Length data is fitted into **Equation 5**, $a = 2$ but $n \neq 0.5$, indicating the leading coefficient of the mathematical relation is 2, but it is not described by a root function.

2. It is discovered that Q factor has a very weak correlation with Pendulum Length when Data is fitted into **Equation 6**.

$$T \simeq aL^n \quad (5)$$

Equation that predicts a “root” relationship between variables Period and Length² when $n=0.5$. Where T (sec) is the period; L (m) is the length of the pendulum wire; a ($[s/m^n]$), n (unitless) are parameters of best fit

$$Q = Q_0(1 + BL + CL^2) \quad (6)$$

Quadratic function that predicts the initial angle to have a quadratic relationship with the period. Where Q (unitless) is the period; Q_0 (unitless), B ($1/m$) and C ($1/m^2$) are parameters for fitting; L (m) is the length of the pendulum.

Section A: Methods and procedures

Below are control and/or preparation methods specifically taken to lower the effects of uncertainties and their reasoning:

1. A **high friction** mat (my leather folder) is placed in between the pendulum and the surface to reduce momentum lost for the mass, effectively decreasing the uncertainty for the period.
2. *Tracker*⁴ (Brown, Christian, Hanson, 2009) is implemented instead of stopwatches or physical protractors to more accurately determine the time for period and angle for amplitude. Lowering the type B uncertainty for both variables as the tracker is as accurate as 30 fps and has a built in rotatable axis system to account for tilted recording angles.
3. No temperature regulators including fans or air conditioning is turned on to ensure no additional net forces are applied to the mass to reduce the predicted random error (type B uncertainty).
4. The string of the pendulum and the camera is ensured to be **straightened and parallel to the wall**, to ensure motion is almost 2 dimensional, this reduces the Type B uncertainty required to account for slight displacements on the 3rd dimension.
5. For subsection aii) and cii) **20 degrees with claimed constant period** was specifically chosen to reduce the impact of uncertainties of period during the initial release angle and period experiment (See subsection ci for more details.)
6. The pendulum’s cable is a thin “fishing wire” tied to a lock of ~42.0g which is attached to a wooden block (See **Figure 5** in **Appendix II**), this is done to fit the lab outline’s requirement of a heavy mass, and a light cable.
7. To eliminate Type A uncertainty, the cable end is taped on the desk so that length is kept constant during motion.

Below is the outline of the steps followed for the lab experiment; it will be split into three subsections, including separate procedures for determining the relationship between initial release angles and the period, the Q factor and the relationship between Period/Q factor and the Length. (Refer to **Figure 5** in the **Appendix II** for my setup.)

Subsection ai): Lab procedures for initial release angle and period

1. Follow the preparation and control methods outlined above and start the recording
2. At a rough marking made for an angle of the pendulum, lift the string and release the mass of the pendulum
3. Allow the mass to complete one (or more) full oscillation then stop the recording.
4. Upload the video onto Tracker, edit the video so that the first frame is the frame of when the mass is released, the last frame is the frame when the mass is momentarily at rest on the same side.
5. Find the period (See **Appendix I**) and actual amplitude (See **Appendix I**) record the data

- Repeat for multiple angles and fit the data onto **Equation 2** then record uncertainties.

Subsection aii): Lab procedures for Q factor

- Follow the preparation methods outlined above and start the recording
- Lift the mass so that it is around 20 degrees (See methods), and release the mass of the pendulum.
- Allow the mass to stay in motion for around one minute and stop the recording.
- Edit the video on Tracker so that the first frame is the frame when the mass is released, the last frame is the frame when the mass is momentarily at rest on the same side after at least 1 minute.
- Track each frame manually (*because my autotrack did not work*) and record a data table of angles, angle against the perpendicular central axis of the pendulum and time.
- Fit the whole set of the data into **Equation 1** with curve.fit⁶. Determine the local maximums (amplitudes) and fit it into **Equation 3**.
- Q factor is determined by two methods,
 - On the exponential decay graph, find the intersection with the line $y = \text{Initial amplitude} \div e$ gives the τ factor, and apply **Equation 4** to find the Q factor.
 - Counting the number of oscillations on the graph of **Equation 1** and find the oscillations it take for amplitude to decay to original amplitude $\times e^{-\pi/2}$ (*because my computer cannot handle large files*), and multiply the count by 2 (See **Figure 6** in **Appendix II**).

Subsection aiii): Lab procedures for Period and Length/Q factor and Length.

Period and Length lab procedures follow the same steps from **Subsection ai Steps 1-5**, with the added step of repeating for multiple pendulum lengths and fitting data into **Equation 5** and record uncertainties.

Similarly, Q factor and Length lab procedures follow the same steps from **Subsection aii Steps 1-5 and 7B**, with the added step of repeating for multiple pendulum lengths and fitting data into **Equation 6** and record uncertainties.

Section B: Graphs and data

Figure 1:

Period of pendulum against the initial release angle function, best fitted into **Equation 2**. Parameters and respective uncertainties are $T_0 = 1.13 \pm 1 \times 10^{-2}(\text{sec})$; $B = -4.16 \times 10^{-4} \pm 5.00 \times 10^{-5}$; $C = 6.59 \times 10^{-3} \pm 1 \times 10^{-3}$. Given vertical uncertainties are ± 0.02 for small angles, a constant period can be concluded for small angles between 0.3 rad for 1.13 ± 0.02 sec. Vertical uncertainties vary with angle. Horizontal uncertainties are too small to be seen ($\pm 4.36 \times 10^{-3}$)

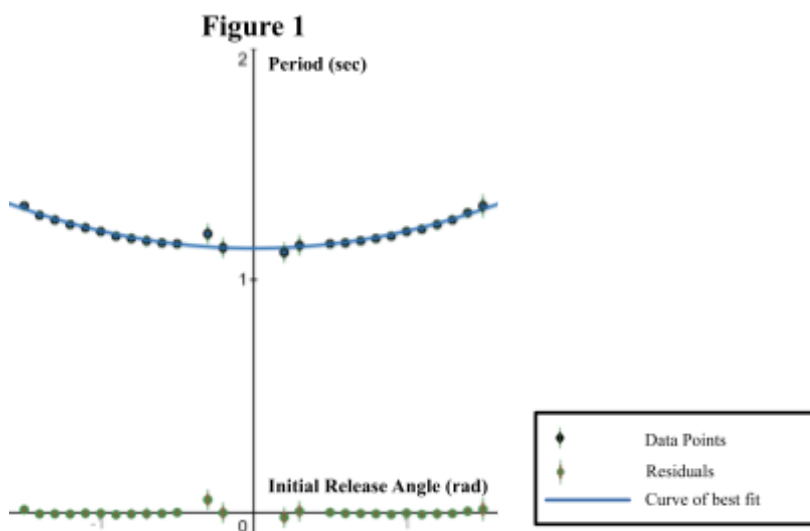


Figure 2:

Amplitude of pendulum against the period function, best fitted to **Equation 3**. Parameters and respective uncertainties are $a = 6.32 \times 10^{11} \pm 5.43 \times 10^9$; $p = 8.65 \times 10^{-3} \pm 7.44 \times 10^{-5}$ (1/sec); $c = 7.89 \times 10^{-2} \pm 6.79 \times 10^{-4}$; $d = -3.30 \times 10^3 \pm 2.84 \times 10^1$ (sec). Intersection with $y = \text{Initial amplitude} \div e \sim 1.20 \times 10^{-1}$ gives the τ factor, giving Q of around 570 ± 30 . Horizontal Uncertainties are too small to be seen ($\pm 7.5 \times 10^{-3}$ sec)

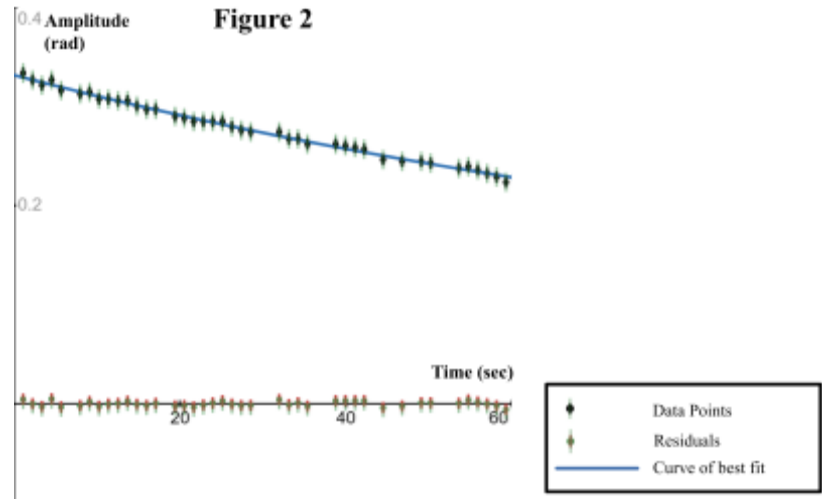


Figure 3:

Period of pendulum against the pendulum length function, best fitted into **Equation 5**. Parameters and respective uncertainties are $a = 2.04 \pm 0.04$ (s/m^{0.457}); $n = 4.57 \times 10^{-1} \pm 0.09 \times 10^{-1}$. This indicates disagreement with the prediction $a = 2$ and $n = 0.5$. Vertical uncertainties vary with angle. Horizontal uncertainties are too small to be seen ($\pm 3 \times 10^{-3}$)

Note a log-log plot is provided in Appendix II, Figure 8

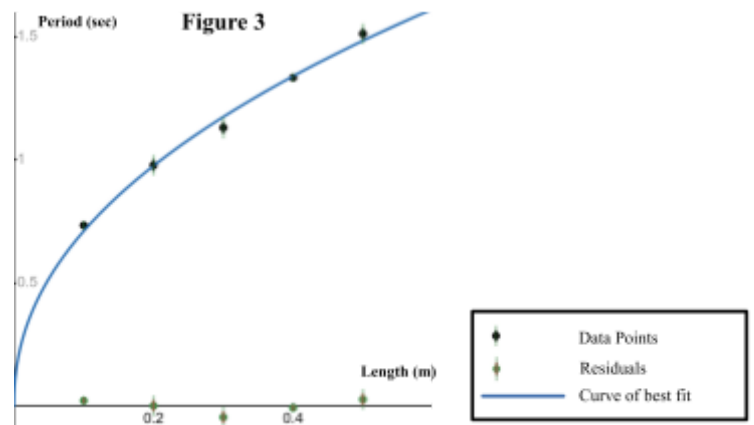
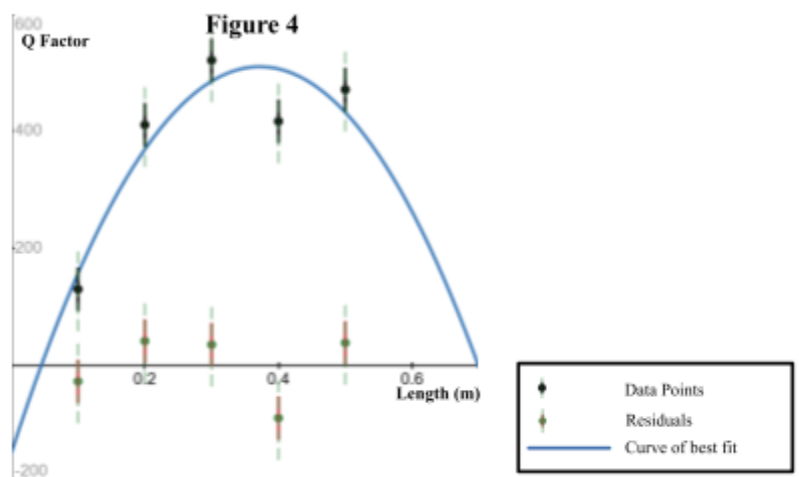


Figure 4:

Q factor of pendulum against the pendulum length function, best fitted into **Equation 6**. Parameters and respective uncertainties are $Q_0 = -1.50 \times 10^2 \pm 0.01 \times 10^2$; $B = -23.3 \pm 17.9$ (1/m); $C = 3.18 \times 10^1 \pm 0.54 \times 10^1$ (1/m²). This indicates a weak quadratic relationship between Q factor and the Length of the pendulum. X-ints are ~ 0.045 m and 0.70 m respectively. The maximum (vertex) of the Q factor = 510 ± 10 described by the best-fitted relation occurs at ~ 0.37 m. Vertical uncertainty is defined to be ± 35 (see **Subsection civ.**) Horizontal uncertainties are too small to be seen ($\pm 3 \times 10^{-3}$)



Section C: Analysis and Uncertainties for Part I

Subsection ci.): Period vs Angle

Largest source of vertical uncertainties in this experiment was the **Type A uncertainty** of each period. Refer to **Figure 1**, the uncertainty was at its highest at small angles 0.2, 0.3 rad, with value of ± 0.02 sec and varies for each individual angle tested. Horizontally, the greatest uncertainty factor was **the manual tracking**, resulting in a range of angles within 1 degree (~ 0.0175 rad) and largest **Type B uncertainty** of $\pm 4.36 \times 10^{-3}$, which is almost neglectable as the wire is almost released at the same spot every time.

Refer to **Figure 1**, collected data is best fitted onto a quadratic series. Experimentally, **B is close to or equal to 0**, as it is always at least 2 times less than the vertical uncertainties at all angles. Graphically, we can deduce that the quadratic curve is **almost symmetric by inspection**. Hence, it is strongly suggested that the pendulum is symmetric in nature.

However, it can be determined that C is not always less than the vertical uncertainty. At angles greater than 0.3 or less than -0.3, C is 1 or 2 factors larger (or occasionally smaller) than the uncertainties at the respective angles; C is **only** almost certainly more than five factors less than the vertical uncertainties at angles between -0.3 and 0.3 with an uncertainty of 0.02 (Refer to **Appendix V** for full values of uncertainties). Hence, it is slightly suggested that the period of the pendulum does in fact **depend on angles**, and is strongly suggested that it is only **approximately constant at small angles** ($<|0.3|$ rad), with the measured average value of 1.13 ± 0.02 sec as suggested in **Equation 7**, as C is more than 5 times less than 0.02 in that range (i.e. at these angles the period is **independent** of the angle)

$$T = 1.13 \pm 0.02, \text{ given } |\text{initial angle}| \leq 0.3 \text{ rad} \quad (7)$$

Subsection cii.): Q factor

Largest uncertainty of time (for both oscillation and decay) was first determined via the **fps of my video on tracker**; 30 fps gives an almost negligible $\pm 8 \times 10^{-3}$ horizontal **Type B** uncertainty for each data point. The largest **vertical Type B** uncertainty for angle is the same as subsection ci, **resultant from the issue of manual tracking** $\pm 4.36 \times 10^{-3}$. (Note: Type A uncertainty is calculated from the standard deviation between the difference between the best fit and data points, but is neglected as it is less than Type B in this case ~ 0.025)

For method A:

Type B uncertainty ~ 6 for τ factor is determined by finding the largest/smallest amplitude and consequently the range for τ to fall in between (i.e between angle $+ 4.36 \times 10^{-3}$ and angle $- 4.36 \times 10^{-3}$). With reference to the uncertainty of the period (± 0.2) in subsection ci, the uncertainty for the Q factor with method A is determined to be $\sim \pm 30$ with **Equation 4**.

For method B:

Type B uncertainty of Q factor is similarly found by estimating the range of numbers of oscillation between largest/smallest amplitude and was **multiplied by 2** (Q/2 was found instead of Q). This gives a Type B uncertainty of $\sim \pm 10$.

Method A gives a τ factor of 206 ± 6 and a Q factor of 570 ± 30 according to **Equation 8**.

Method B gives a Q factor of 620 ± 10 according to **Equation 9**.

$$Q_1 = 570 \pm 30 \quad (8)$$

$$Q_2 = 620 \pm 10 \quad (9)$$

Using Python, two separate bell curves as far as 3 times the uncertainty is plotted (Refer to **Figure 7** in **Appendix II**). This gives a **8.83% chance** that the Q factor from Method A is equal to the Q factor from Method B. This chance is **close to 5%** (i.e occurrence chance for values of 2 standard deviations), hence it is suggested that my data slightly disagrees with the prediction that both Q values are equal for the two methods. This is represented by **Equation 10**

$$Q_1 \neq Q_2 \quad (10)$$

Subsection ciii.): Period vs Length

Vertical uncertainty sources vary (Refer to **Appendix V**). For lengths 0.2, 0.3 and 0.5m, Type A uncertainties of each period were the greatest source of uncertainty; For lengths 0.1m and 0.4m, Type B uncertainty due to the 30 fps of the video was the greatest ($\sim \pm 8 \times 10^{-3}$). Horizontally, Type A uncertainty is ignored as string wire is measured every time to be constant. Subsequently, a negligible Type B uncertainty of the range of measurement of my measuring tape is determined to be $\sim 3 \times 10^{-3}$.

Refer to **Figure 3**, when data is best fitted onto a power law equation, it is determined that parameter $a = 2.04$ is within 1 uncertainty (0.04) from 2 and that parameter $n = 0.456$ is 5 uncertainties (0.009) from 0.5. Since the parameter a is within the prediction and the parameter n is significantly off from the prediction, it is strongly suggested that the relationship between Period and Length is not described by a root function and that the leading coefficient (a) that describes the relation is 2.00.

$$\text{For } T = aL^n, a \approx 2, n \neq 0.5 \quad (11)$$

Subsection civ.): Q factor vs Length

With the method of determining Q factor defined in **Section D, Part I conclusions**, Vertical **Type B** uncertainties are defined to be the issue caused by the measurement from the stopwatch. The reaction time of me is defined to be around 1 second, since each trial for Q factor ran for ~ 70 seconds, and the final Q factor found is multiplied by 2, the range/4 uncertainty would be a resultant ± 35 . (This is taken instead of Type A uncertainty as the value is greater) Horizontal uncertainty follows the same procedures in **Subsection ciii.)**, where a small and negligible Type B uncertainty is determined to be $\sim 3 \times 10^{-3}$.

With trial and error, it is determined that fitting data into a Quadratic relationship best relates the Q factor to the length of my pendulum. Refer to **Figure 4**, parameters Q_0 , B and C are all greater than their uncertainties respectively, indicating a suitable fit. That said, when the uncertainties are expressed in the form of percentages, they are 0.32%, 17.07% and 75.56% respectively. For parameters B and C, these values are too large and no strong conclusions can be established. Specifically, a 75% percentage uncertainty shows an extremely weak correlation. (or no relation at all.) This fit is best described by **Relation 12**.

$$Q \approx -150(31.8L^2 - 23.7L + 1), \text{ for } 0.045m \leq L \leq 0.700m \quad (12)$$

With this fit, the Q factor is only meaningful between lengths of 0.045-0.700m (x-ints), anything above and below gives negative Q factors. This relation indicates that Q factor increases from ~ 0.045 -0.37m and decreases from ~ 0.37 -0.70m. It is determined via the vertex that Q factor peaks at ~ 0.37 m at 510 ± 10 .

Section D: Conclusions

In conclusion, my data in Part I suggests the following:

1. It is strongly suggested that the pendulum is symmetric.
2. It is slightly suggested that the period is dependent on the angle.
3. It is strongly suggested that the period is around constant at a positive angle less than or equal to 0.3 rad.
4. It is slightly suggested that both methods of Q factor produce different results.

With the above conclusions, in the second part of the lab related to the length of the pendulum and the Q factor, two measurements (and their justification) will be taken to favor the quality of the data:

1. The initial angle is to be **less than or equal to 0.3 rad** to ensure the period is **independent** of the angle and kept constant.
2. Q factor is to be measured by **method B** as it produces a smaller uncertainty and requires less time to operate, i.e. Q factor is defined (in subsequent parts of the lab) to be the 2 times the amount of oscillations it takes for the original amplitude to decrease to original amplitude $\times e^{-\pi/2}$

In conclusion, my data in Part II suggests the following:

1. It is strongly suggested that the relationship between the Period and the Length is not described by a root function
2. It is strongly suggested that the leading coefficient that describes the above relationship is 2.
3. It is suggested that an extremely weak quadratic correlation that describes the relationship between the Q factor and Length of the pendulum exists, with Q factor maximizing at a value of 510 ± 10 at $\sim 0.37\text{m}$.

Section E: Appendix

i.) Definitions:

Period - The period of a pendulum is defined to be the time it takes for a pendulum to complete one full oscillation⁵. In Tracker, the period of one oscillation is defined to be the time difference between the edited last frame of the video and the edited first frame of the video. (OpenStax, 2016)

Amplitude - The amplitude of a pendulum is defined to be the maximum angle it reaches during each oscillation⁵. In Tracker, the amplitude is the angle between the central axis in the middle of the pendulum and the string when the mass is momentarily stationary at its maximum height. The central axis can be found using Tracker's rotatable origin system. (OpenStax, 2016)

ii.) Additional figures:

Figure 5

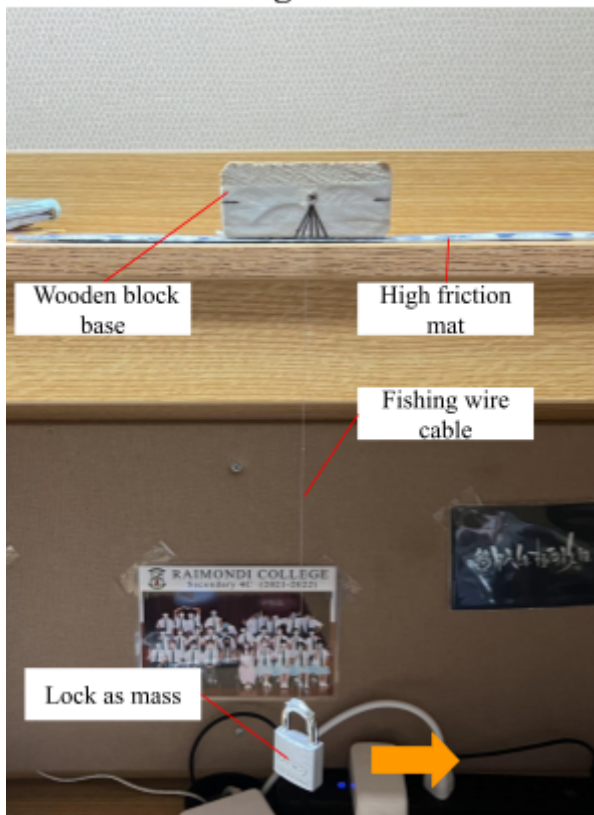


Figure 5:
Direction of which the Q factor is measured is indicated by the orange arrow parallel to the video angle.

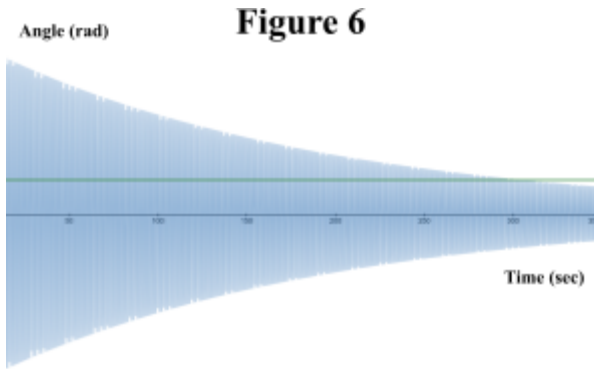


Figure 6:

Blue line: Motion of pendulum against time function. Damped Oscillator best fit in the form $ae^{-bx}\cos(\pi/2 - cx + d) + f$, with parameters $a = 3.23 \times 10^{-1}$, $b = 5.27 \times 10^{-3}$, $c = -5.51$, $d = -1.48$, $f = 2.50 \times 10^{-3}$. Vertical uncertainties are $\pm 4.36 \times 10^{-3}$.

Green line: Horizontal line $y = \text{Initial amplitude } (e^{-\pi/2}) \sim 6.77 \times 10^{-2}$.

Counting Oscillation until horizontal line is above amplitude gives Q of around **620** ± 10

Horizontal Uncertainties are too small to be seen (7.5×10^{-3})

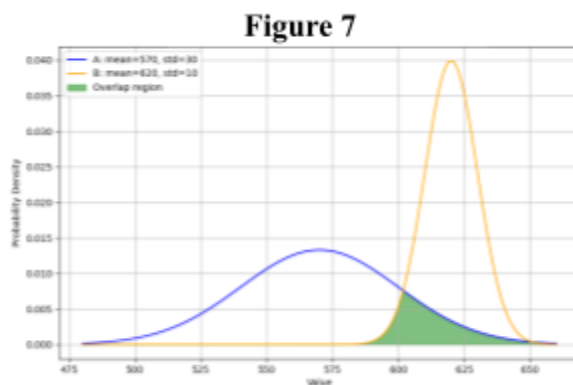


Figure 7:

Bell Curves of two values of Q ,

Blue Curve: Q_1

Yellow curve: Q_2

Green overlapping area $\sim 8.83\%$ gives probability of equal values.

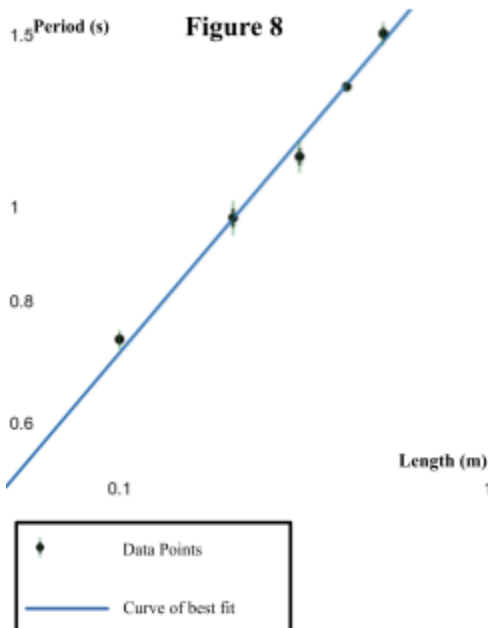


Figure 8:

Log-log plot equivalent of **Figure 3**,

Period of pendulum against the pendulum length

function, best fitted into **Equation 5**. Parameters and respective uncertainties are $a = 2.04 \pm 0.04 \text{ (s/m}^{0.457}\text{)}$; $n = 4.57 \times 10^{-1} \pm 0.09 \times 10^{-1}$. This indicates disagreement with the prediction $a = 2$ and $n = 0.5$. Vertical uncertainties vary with angle. Horizontal uncertainties are too small to be seen ($\pm 3 \times 10^{-3}$)

iii.) AI Application in this report:

In this report, AI was applied 6 times all for the purpose of writing up code:

In Part I:

Two of these times was to write a python code to determine the % uncertainty for the fit between my parameters and the curve of best fit for both the Amplitude/Time exponential and Period/Initial Release Angle quadratic curve. The other instance AI was applied was to write a Python code to generate Figure 7 and determine the overlapping area between the bell curves. AI was also applied to determine the standard deviation of the difference between the Amplitude/Time exponential best-fit curve and the individual data points.

In Part II:

AI was applied twice to write a python code to determine the % uncertainty for the fit between my parameters and the curve of best fit for both the power law function of Period/Length data set and the quadratic function of Q factor/Length data set

This is the link for all the codes utilized: [All codes used for Report 2](#)

iv.) Use of curve.fit and Desmos.com for Section B:

All of the equations were best fitted using curve.fit⁶ (Boggs, Byrd, Rogers, Schnabel, 1992).

All of the graphs (except Figure 7) were plotted using Desmos.com⁷

v.) Tables of varying vertical uncertainties:

This is the link to all non-constant vertical uncertainties:

[Non-constant Vertical Uncertainties for Report 2](#)

vi.) Citations:

1. The Editors of Encyclopaedia Britannica. (2008, November 19). *Simple harmonic motion*. Encyclopædia Britannica. <https://www.britannica.com/science/simple-harmonic-motion>
2. Wilson, Brian, "PHY180 Lab Project", downloaded from Quercus, 2024.
3. Massachusetts Institute of Technology. (2017). *Res.8-009 (Summer 2017), lecture 4: Damped oscillations*. Lecture 04: Damped Oscillations. https://ocw.mit.edu/courses/res-8-009-introduction-to-oscillations-and-waves-summer-2017/mitres_8_009su17_lec4.pdf
4. Brown, D., Christian, W., & Hanson, R. M. (2009, December). TrackerVersion (6). *Tracker Video Analysis and Modelling Tool*. Retrieved October 3, 2024, from <https://tracker.physlets.org/>.
5. Moebis, W., Ling, S. J., & Sanny, J. (2016, September 19). *15.1 simple harmonic motion - university physics volume 1*. OpenStax. <https://openstax.org/books/university-physics-volume-1/pages/15-1-simple-harmonic-motion#:~:text=List%20the%20characteristics%20of%20simple%20harmonic%20motion;%20Explain%20the%20concept>
6. Boggs, P., Byrd, R., Rogers, J., & Schnabel, R. (1992). *Curve fitting with X and Y Uncertainties*. Curve.fit. <https://curve.fit/>
7. Desmos Studio, PBC. (n.d.). *Beautiful free math*. Desmos. <https://www.desmos.com/>